

TIC 42058853

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.730779 d
Epoch (t0) 1683.8078 BTJD
Duration 0.2440 d
Depth 1.314 % (13,138 ppm)
SNR 160.5

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 247 (gate ≥ 3)
Robust depth ratio 0.89 (gate ≥ 0.50)
Robust depth sigma 135.9 (gate ≥ 7.0)
Odd/even z-score 1.91 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

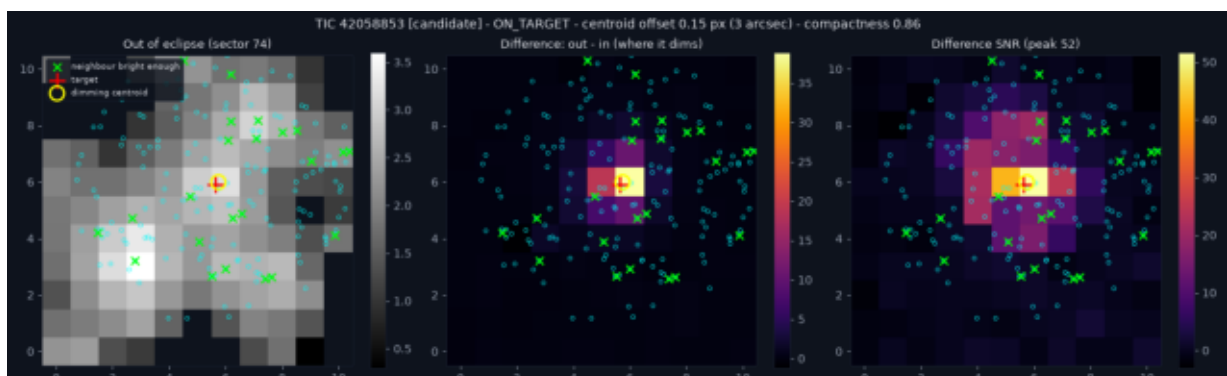
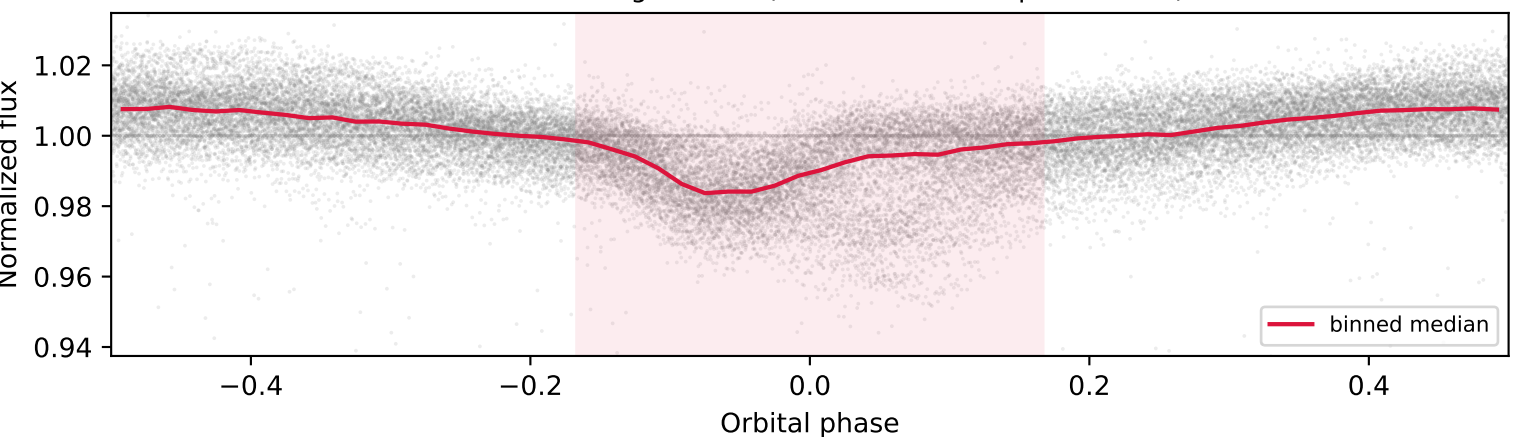
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 74 light-curve dip in this sector: 77.6 sigma t0 refined by -0.0400 phase
Difference-image peak SNR 51.73 (gate ≥ 5.0) Compactness 0.86 (gate ≥ 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 42058846
1.12 px away, dTmag +3.15 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.